Now we consider the following example.

Example

Solve the equation

$$y'' - 3y' - 4y = 2e^{-x}.$$

Solution

We have

$$P(t) = t^2 - 3t - 4 = 0 \Longrightarrow t_1 = -1, t_2 = 4.$$

Then the complementary function is

$$y_c(x) = C_1 e^{-x} + C_2 e^{4x},$$

where C_1, C_2 are arbitrary constants.

We find a particular solution of the equation

$$y'' - 3y' - 4y = 2e^{-x}.$$

Following the previous example, we choose

$$y_p(x) = Ae^{-x}.$$

Substituting this function into the given equation, we obtain

$$(A + 3A - 4A)e^{-x} = 2e^{-x} \quad \Longleftrightarrow \quad 0 \cdot e^{-x} = 2e^{-x}.$$

The last equation gives no A, and hence there is no particular solution of the form above.

What happened?

The reason is that the function $y(x) = e^{-x}$ is one of the two solutions of the homogeneous equation.
Then we have to use the so-called modification rule.

(a') **Modification rule:** If a term in our choice for y_p happens to be a solution of the associated homogeneous equation, then multiply this term by x (or by x^2 if this solution corresponds to a double root of the characteristic equation of the associated homogeneous equation).

Choose a particular solution in the form

$$y_p(x) = Axe^{-x},$$

and substitute it into the non-homogeneous equation, we obtain

$$-5Ae^{-x} + 0xe^{-x} = 2e^{-x},$$

which gives $A = -\frac{2}{5}$, and so $y_p(x) = -\frac{2}{5}xe^{-x}$.

Thus the general solution of the given non-homogeneous equation is

$$y(x) = y_c(x) + y_p(x) = C_1 e^{-x} + C_2 e^{4x} - \frac{2}{5} x e^{-x},$$

where C_1, C_2 are arbitrary constants.

□

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1. Some Simple Forms of Second-Order Equations
2. Second-order Linear Differential Equations
3. Homogeneous Lin. Diff. Eq. with Constant-Coefficients
4. The Method of Undetermined Coefficients
5. The Method of Variation-of-Parameters
6. Series Solutions to Linear Differential Equations

The Method of Variation-of-Parameters

The aim of this section is to determine **particular solutions** to non-homogeneous differential equations, provided that the **general solution** to the associated homogeneous differential equation is **known**.

This **variation-of-parameters** method is **not** restricted to differential equations with constant coefficients.

Consider the **second-order** linear differential equation

$$y'' + a_1 y' + a_2 y = F, \tag{1}$$

where a_1, a_2, and F are **continuous** on an interval I.

Suppose that $y_1(x)$ and $y_2(x)$ are two linearly independent solutions to the associated homogeneous equation

$$y'' + a_1 y' + a_2 y = 0, \tag{2}$$

on I, so that the general solution to Equation (2) on I is

$$y_c = C_1 y_1(x) + C_2 y_2(x).$$

Discussion

The **variation-of-parameters** method consists of **replacing** the **constants** C_1, C_2 by **functions** $C_1(x), C_2(x)$ (that is, allow the parameters C_1, C_2 to vary), determined in such a way that the resulting function

$$y_p = C_1(x)y_1(x) + C_2(x)y_2(x)$$

is a particular solution to Equation (1).

We have the following result.

Theorem

Consider a differential equation

$$y'' + a_1 y' + a_2 y = F, \tag{3}$$

where a_1, a_2, and F are (at least) continuous on an interval I.

Let y_1, y_2 be **linearly independent solutions** *to the associated homogeneous equation*

$$y'' + a_1 y' + a_2 y = 0 \tag{4}$$

on I.

Theorem (continued)

Then a **particular solution** *to Equation (3) is*

$$y_p(x) = C_1(x)y_1(x) + C_2(x)y_2(x), \tag{5}$$

where C_1, C_2 *satisfy* $\begin{cases} y_1 C_1' + y_2 C_2' = 0 \\ y_1' C_1' + y_2' C_2' = F. \end{cases}$

Proof (optional)

- First we assume to have $C_1(x), C_2(x)$ in Equation (5) with two conditions said above.

From this it follows that

$$y_p' = C_1' y_1 + C_1 y_1' + C_2' y_2 + C_2 y_2' = C_1 y_1' + C_2 y_2',$$

due to the first condition for C_1, C_2. Hence

$$y_p'' = C_1' y_1' + C_1 y_1'' + C_2' y_2' + C_2 y_2'' = C_1 y_1'' + C_2 y_2'' + F,$$

due to the second condition for C_1, C_2.

Substituting y_p, y_p', and y_p'' into the LHS of Equation (3) yields

$$\begin{aligned} y_p'' + a_1 y_p' + a_2 y_p &= (C_1 y_1'' + C_2 y_2'' + F) \\ &\quad + a_1(C_1 y_1' + C_2 y_2') + a_2(C_1 y_1 + C_2 y_2) \\ &= C_1(y_1'' + a_1 y_1' + a_2 y_1) \\ &\quad + C_2(y_2'' + a_1 y_2' + a_2 y_2) + F \\ &= F, \end{aligned}$$

as y_1, y_2 are (linearly independent) solutions to Equation (4).

- The rest is to show that $C_1(x), C_2(x)$ do really **exist**.

Indeed, the system

$$\begin{cases} y_1 C_1' + y_2 C_2' = 0 \\ y_1' C_1' + y_2' C_2' = F \end{cases}$$

has a matrix of coefficients with the determinant

$$\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix},$$

which is **non-zero** on I (because y_1, y_2 are **linearly independent** on I), and hence the system, by **Cramer's rule**, has a unique solution for C_1', C_2', from which we obtain C_1, C_2.

Example

Solve $y'' + y = \sec x$.

Solution

Since two linearly independent solutions to the associated homogeneous differential equation are $y_1 = \cos x, y_2 = \sin x$, a particular solution to the given differential equation is

$$y_p = C_1(x)y_1 + C_2(x)y_2 = C_1(x)\cos x + C_2(x)\sin x, \tag{6}$$

where C_1, C_2 satisfy

$$\begin{cases} \cos x \; C_1' + \sin x \; C_2' = 0 \\ -\sin x \; C_1' + \cos x \; C_2' = \sec x. \end{cases}$$

By Cramer's rule, the solution to this system is

$$C_1' = -\sin x \sec x, C_2' = \cos x \sec x = 1.$$

Consequently,

$$C_1(x) = -\int \sin x \sec x \, dx = -\int \frac{\sin x}{\cos x} \, dx = \log|\cos x|$$

and

$$C_2(x) = \int 1 \, dx = x.$$

(here we have set the integration constants to zero, since we require only **one** particular solution).

Substituting into Equation (6) we have

$$y_p = \cos x \log|\cos x| + x \sin x$$

so that the general solution to the given differential equation is

$$y(x) = \alpha_1 \cos x + \alpha_2 \sin x + \cos x \log|\cos x| + x \sin x,$$

where α_1, α_2 are arbitrary constants.

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Finally, we consider the **possibility** of representing **solutions** to linear differential equations in the form of some types of (infinite) **series**.

• We first review basic properties of **power series**, that is, the series of the form

$$\sum_{n=0}^{\infty} a_n (x-a)^n,$$

where a_n and a are constants.

• W.l.o.g. we can assume that $a = 0$, (why ? - because the substitution $z = x - a$ has the effect of transforming the power above to $\sum_{n=0}^{\infty} a_n z^n$).

Theorem

For the series

$$\sum_{n=0}^{\infty} a_n x^n$$

one of the following is true:

1) *The series converges only at* $x = 0$.
2) *The series converges for all real* x.
3) *There exists a positive real number* R *such that the series converges (absolutely) for* $|x| < R$ *and diverges for* $|x| > R$.

• The number R is called the **radius of convergence** and $(-R, R)$ is called the **interval of convergence** of the power series.

• The convergence or divergence of the series at the **endpoints** $x = \pm R$ must be investigated **separately**.

• For **finding** the radius of convergence R of a power series, we can use the following result.

Theorem (**Radius of convergence**)

Suppose for $\sum_{n=0}^{\infty} a_n x^n$ *we have*

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \alpha.$$

1) *If* $\alpha = 0$, *then* $R = \infty$, *i.e. the series converges for all* x.
2) *If* $\alpha = \infty$, *then* $R = 0$, *i.e. the series converges only at* $x = 0$.
3) *If* $\alpha \in (0, \infty)$, *then*

$$R = \frac{1}{\alpha} = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right| \quad \textit{(Cauchy-Hadamard formula)}.$$

Example

Determine a radius of convergence of the following power series.

1) $\sum_{n=1}^{\infty} n^n x^n$.

2) $\sum_{n=1}^{\infty} \frac{1}{n!} x^n$.

3) $\sum_{n=1}^{\infty} \frac{1}{2^n n^2} x^n$.

Solution

1) For $\sum_{n=1}^{\infty} n^n x^n$, we have

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+1)^{n+1}}{n^n} = \lim_{n\to\infty}\left\{\frac{(n+1)^n(n+1)}{n^n}\right\}$$

$$= \lim_{n\to\infty}\left\{\left(1+\frac{1}{n}\right)^n\cdot(n+1)\right\}$$

$$= \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n\cdot\lim_{n\to\infty}(n+1) = \infty$$

and hence, $R = 0$.

2) Next, for $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n!} x^n$, we have

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{\dfrac{1}{(n+1)!}}{\dfrac{1}{n!}}$$

$$= \lim_{n\to\infty} \frac{n!}{(n+1)!} = \lim_{n\to\infty} \frac{1}{n+1} = 0$$

and hence, $R = \infty$.

3) Finally, for $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n n^2} x^n$, we have

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \frac{\dfrac{1}{2^{n+1}(n+1)^2}}{\dfrac{1}{2^n n^2}}$$

$$= \lim_{n\to\infty} \frac{2^n n^2}{2^{n+1}(n+1)^2} = \lim_{n\to\infty} \frac{n^2}{2(n+1)^2} = \frac{1}{2}$$

and hence, $R = 2$.

At the endpoints $x = \pm 2$: **Exercise !**

Power series can be differentiated (and also integrated - but it is not needed in this chapter) by the following result.

Differentiation of power series

Suppose that

$$\sum_{n=0}^{\infty} a_n x^n$$

has radius of convergence R, and let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad |x| < R.$$

Then $f(x)$ can be differentiated an arbitrary number of times on the interval $|x| < R$.

Furthermore, the derivatives can be obtained by termwise differentiation. Thus,

$$f'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$f''(x) = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

and so on for higher-order derivatives.

Definition

A function is said to be **analytic** at $x = a$, if it can be represented in the form of a convergent power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ with nonzero radius of convergence.

- Many of the familiar elementary functions are analytic at all points.
- Note also that if functions $f(x)$ and $g(x)$ are analytic at $x = a$, then so also are $f(x) \pm g(x)$, $f(x)g(x)$ and $\dfrac{f(x)}{g(x)}$ (provided that $g(a) \neq 0$).

Definition

- The point $x = a$ is called an **ordinary point** of the differential equation

$$y'' + p(x)y' + q(x)y = 0,$$

if p and q are both analytic at $x = a$.

- Any point which is not an ordinary point is called **singular point** of the equation.

The **solution technique** is given in the following result.

Theorem

Let p and q be analytic at $x = a$, and suppose that their power series expansions are valid for $|x - a| < R$.

- *Then the general solution to the differential equation*

$$y'' + p(x)y' + q(x)y = 0$$

can be represented in the form of a power series

$$y(x) = \sum_{n=0}^{\infty} a_n (x - a)^n,$$

with radius of convergence at least R.

Theorem (continued)

- *The coefficients in this series solution can be determined in terms of a_0 and a_1 by directly substituting $y(x)$ into the given equation.*
- *The resulting solution is of the form*

$$y(x) = a_0 y_1(x) + a_1 y_2(x),$$

where y_1 and y_2 are linearly independent solutions to the given equation on the interval of existence.

- *If the initial conditions $y(a) = A, y'(a) = B$ are imposed, then $a_0 = A, a_1 = B$.*

We illustrate the use of this solution technique with some examples.

Example

Show that the equation

$$(1 + x^2)y'' + 3xy' + y = 0$$

has two linearly independent series solutions centered at $x = 0$, and determine a lower bound on the radius of convergence of these series.

Solution

In the standard form, the given equation is written as

$$y'' + \frac{3x}{1+x^2}y' + \frac{1}{1+x^2}y = 0.$$

From this equation we can conclude that $x = 0$ is an ordinary point.

Then the given equation does have two linearly independent series solutions centered at $x = 0$. In this case, since

$$p(x) = \frac{3x}{1+x^2} \text{ and } q(x) = \frac{1}{1+x^2},$$

with radii of convergence R_1 and R_2 respectively, the radius of convergence R of the power series solutions, by the solution technique theorem, will be at least $\min\{R_1, R_2\}$.

We can **check (!)** that $R_1 = R_2 = 1$, and so a lower bound on the radius of convergence of the power series solutions to the given differential equation is also $R = 1$.

Example

Determine two linearly independent series solutions in power series of x to the equation

$$y'' - 2xy' - 4y = 0,$$

and find the radius of convergence of these series.

Solution

The point $x = 0$ is an ordinary point of the given differential equation, and therefore the theorem above can be applied with $a = 0$.

In this case, we have

$$p(x) = -2x, \quad q(x) = -4.$$

Note that both p and q are polynomials, and hence their power series expansions about $x = 0$ are valid for all x.

Then the general solution to the equation can be represented in the form

$$y(x) = \sum_{n=0}^{\infty} a_n x^n,$$

and this power series solution will converge for all x.

Differentiating this y, we obtain

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Substituting into the given equation $y'' - 2xy' - 4y = 0$ yields

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 2\sum_{n=1}^{\infty} n a_n x^n - 4\sum_{n=0}^{\infty} a_n x^n = 0,$$

or equivalently,

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1)a_{n+2} - 2n a_n - 4a_n\right] x^n = 0.$$

From this it follows that

$$(n+2)(n+1)a_{n+2} - 2na_n - 4a_n = 0$$

for $n = 0, 1, \ldots$, which is written in the form

$$a_{n+2} = \frac{2}{n+1} a_n, \ n = 0, 1, \ldots$$

This shows that all of the even coefficients can be expressed in terms of a_0, whereas all of the odd coefficients can be expressed in terms of a_1.

We are going to determine the exact form of these coefficients.

Case 1: n is even.

$$\text{For } n = 0: \quad a_2 = 2a_0.$$

$$\text{For } n = 2: \quad a_4 = \frac{2}{3}a_2 = \frac{2^2}{1 \cdot 3}a_0.$$

$$\text{For } n = 4: \quad a_6 = \frac{2}{5}a_4 = \frac{2^3}{1 \cdot 3 \cdot 5}a_0.$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots$$

The general even term is

$$a_{2n} = \frac{2^n}{1 \cdot 3 \cdot 5 \cdots (2n-1)}a_0, \ n = 1, 2, \ldots.$$

Case 2: n is odd.

$$\begin{aligned}
&\text{For } n=1: \quad a_3 = a_1.\\
&\text{For } n=3: \quad a_5 = \frac{2}{4}a_3 = \frac{1}{1\cdot 2}a_1.\\
&\text{For } n=5: \quad a_7 = \frac{2}{6}a_5 = \frac{1}{1\cdot 2\cdot 3}a_1.\\
&\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots
\end{aligned}$$

The general odd term is

$$a_{2n+1} = \frac{1}{1\cdot 2\cdot 3\cdots n}a_1 = \frac{1}{n!}a_1,\ n=1,2,\ldots.$$

Thus we obtain

$$y(x) = a_0 \left[1 + \sum_{n=1}^{\infty} \frac{2^n}{1 \cdot 3 \cdots (2n-1)} x^{2n} \right] + a_1 \sum_{n=0}^{\infty} \frac{1}{n!} x^{2n+1}.$$

Consequently, by the solution technique theorem, two linearly independent solutions to the equation on $(-\infty, \infty)$ are

$$y_1(x) = 1 + \sum_{n=1}^{\infty} \frac{2^n}{1 \cdot 3 \cdots (2n-1)} x^{2n}, \quad y_2(x) = \sum_{n=0}^{\infty} \frac{1}{n!} x^{2n+1}.$$

□

There are several linear differential equations that arise frequently in engineering and whose solutions can be obtained only by using a power series technique.

Among the most important of these are the following:

1) **Legendre equation** $(1-x^2)y''-2xy'+\alpha(\alpha+1)y=0$,
2) **Hermite equation** $y''-2xy'+2\alpha y=0$,
3) **Chebyshev equation** $(1-x^2)y''-xy'+\alpha^2 y=0$,

where α is an arbitrary constant.

Since $x = 0$ is an ordinary point of these equations, we can obtain a series solution in powers of x.

(For those who are interested in this topic, please see (different) references !)

END OF CHAPTER 4